

Binary Subsequences

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to find a minimum length bit string that has exactly n distinct subsequences.

For example, a correct solution for $n = 6$ is `101` whose distinct subsequences are `0`, `1`, `01`, `10`, `11` and `101`.

Input

The only input line has an integer n .

Output

Print one bit string: a solution to the task. You can print any valid solution.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
6	101